Lossless compression: Data compression technique that exactly reconstructs the original data, such as LZW compression.

Lossy compression: Data compression methods that only reconstruct an approximation to the original data. This allows higher compression ratios to be achieved. JPEG is an example.

Matched filtering: Method used to determine where, or if, a know pattern occurs in a signal. Matched filtering is based on correlation, but implemented by convolution.

Mathematical equivalence: A way of using complex numbers to represent real problems. Based on Euler's relation equating sinusoids with complex exponentials. See *substitution* for comparison.

Mean: The average value of a signal or other group of data.

Memoryless: Systems where the current value of the output depends only on the current value of the input, and not past values.

MFLOPS: Million-Floating-Point-Operations-Per-Second; a common way of expressing computer speed. See *MIPS* for comparison.

MIPS: Million-Instructions-Per-Second; a common way of expressing computer speed. See *MFLOPS* for comparison.

Mixed signal: Integrated circuits that contain both analog and digital electronics, such as an ADC placed on a Digital Signal Processor.

Modulation transfer function (MTF): Imaging jargon for the *frequency response*.

Morphing: Gradually warping an image from one form to another. Used for special effects, such as a man turning into a werewolf.

Morphological: Usually refers to simple non-linear operations performed on binary images, such as erosion and dilation.

Moving average filter: Each sample in the output signal is the average of many adjacent samples in the input signal. Can be carried out by convolution or recursion.

MPEG: Compression standard for video, such as digital television.

Mu law: Companding standard used in the United States. Allows digital voice signals to be represented with only 8 bits instead of 12 bits by making the quantization levels unequal. See "*A*" *law* for comparison.

Multiplexing: Combining two or move signals together for transmission. This can be carried out in many different ways.

Multirate: Systems that use more than one sampling rate. Often used in ADC and DAC to obtain better performance, while using less electronics.

Natural frequency: A frequency expressed in radians per second, as compared to cycles per second (hertz). To convert frequency (in hertz) to natural frequency, multiply by 2π.

Negative frequencies: Sinusoids can be written as a positive frequency: $\cos(\omega t)$, or a negative frequency: $\cos(-\omega t)$. Negative frequencies are included in the complex Fourier transform, making it more powerful.

Normal distribution: A bell shaped curve of the form: e^{x^2}. Also called a *Gaussian*.

NTSC: Television standard used in the United States, Japan, and other countries. See *PAL* and *SECAM* for comparison.

Nyquist frequency, Nyquist rate: These terms refer to the sampling theorem, but are used in different ways by different authors. They can be used to mean four different things: the highest frequency contained in a signal, twice this frequency, the sampling rate, or one-half the sampling rate.

Octave: A factor of two in frequency.

Odd order filter: An analog or digital filter having an odd number of poles.

Opening: A morphological operation defined as a dilation operation followed by an erosion operation.

Optimal filter: A filter that is "best" in some specific way. For example, Wiener filters produce an optimal signal-to-noise ratio and matched filters are optimal for target detection.

Overlap add: Method used to break long signals into segments for processing.

PAL: Television standard used in Europe. See *NTSC* for comparison.